📘 Kubernetes Glossary

Term	What it Means	Simple Explanation / Example
🌐 Kubernetes	Container orchestration platform	Manages containers automatically (deploy, scale, heal)
🏗️ Cluster	Group of machines	One Kubernetes setup = multiple servers
🖥️ Node	Worker machine	A server where pods run
🧠 Control Plane	Brain of cluster	Decides *what runs where*
⚙️ Worker Node	Execution machine	Runs application containers
📦 Pod	Smallest deployable unit	One or more containers running together
🐳 Container	App runtime	Runs your app (Docker container)
🧾 Image	App blueprint	Used to create containers
🧩 Namespace	Logical isolation	Separate environments like `dev`, `prod`
🚀 Deployment	App manager	Ensures correct number of pods
🔢 ReplicaSet	Pod counter	Keeps desired pod count
🔗 Service	Network access	Exposes pods internally or externally
🏠 ClusterIP	Internal service	Access only inside cluster
🌍 NodePort	External service	Exposes app using node IP + port
☁️ LoadBalancer	Cloud service	Exposes app with public IP
🌐 Ingress	HTTP routing	Routes traffic using domain/path
🚪 Ingress Controller	Traffic handler	Implements ingress rules
⚙️ ConfigMap	Non-secret config	Stores env variables, configs
🔐 Secret	Sensitive data	Stores passwords, tokens
💾 Volume	Storage	Data shared with containers
📦 PersistentVolume (PV)	Actual storage	Disk provided by admin/cloud (storage volume that is used to persist application data.)

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📄 PersistentVolumeClaim (PVC)	Storage request	Pod asks for storage (request for storage)
🗄️ StatefulSet	Stateful apps	For DBs like MySQL
📡 DaemonSet	One pod per node	Log agents, monitoring
🧪 Job	One-time task	Runs once (backup, batch job)
⏰ CronJob	Scheduled job	Runs jobs at fixed time
🏷️ Label	Key-value tag	Used to select objects
🎯 Selector	Matching rule	Connects service to pods
⚠️ Taint	Node restriction	Prevents pod scheduling
✅ Toleration	Taint override	Allows pod on tainted node
📍 Affinity	Pod placement	Controls where pods run (✅ “Place this Pod here / near this.”)
🚫 Pod Anti-Affinity	Separation rule (improves availability)	Spread pods across nodes (❌ “Do NOT place this Pod on the same node (or zone))
📈 HPA	Auto scaling	Scales pods based on CPU
🔄 Rolling Update	Zero downtime deploy	Updates pods gradually
🧑‍💻 kubectl	CLI tool	Command line for Kubernetes
🗄️ etcd	Cluster database	Stores cluster state
🎯 Scheduler	Pod placer	Chooses node for pod
🔧 Controller Manager	Maintainer	Keeps desired state
🔌 API Server	Entry point	All requests go through this
📦 Helm	Package manager	Installs apps using charts
📊 Chart	Helm package	Predefined Kubernetes resources
🧩 CRD	Custom resource	Add your own resource type
🤖 Operator	Smart controller	Manages complex apps
❤️ Probe (Liveness)	Health check	Restarts unhealthy pod
🚦 Probe (Readiness)	Traffic check	Controls traffic flow
🚀 Probe (Startup)	Boot check	For slow-starting apps

📗 Advanced Kubernetes Glossary

Term	What it Is	Simple Explanation / Real Use
🌐 Kubernetes API Group	Resource category (API groups make it easier to extend the Kubernetes API) to solve specific user needs.	Example: `apps/v1` , `batch/v1`
🧾 API Version	Resource version	Example: `apps/v1` for Deployment
📄 Manifest	YAML file	Describes Kubernetes objects
🎯 Desired State	Target configuration	What you *want* Kubernetes to maintain
📊 Current State	Actual condition	What is currently running
🔄 Reconciliation Loop	Continuous check	K8s keeps fixing differences
🚦 Admission Controller	Request validator	Allows or denies API requests
✏️ Mutating Webhook	Modifies requests	Adds labels automatically
✅ Validating Webhook	Validates rules	Blocks wrong configs
🌐 CNI	Networking plugin	Calico, Flannel
💾 CSI	(Container Storage Interface) Storage plugin	Connects storage systems
🔗 kube-proxy	Network manager	Enables service networking
📍 Endpoint	Pod IP list	Actual backend pods
📦 EndpointSlice	Scalable endpoints	Improved endpoint handling
🧠 Service Mesh	Traffic control	Istio, Linkerd
🔄 Sidecar	Helper container	Logging, proxy container
🚀 Init Container	Pre-run container	Runs before app starts
⚠️ Eviction	Pod removal	Due to memory/disk pressure
💥 OOMKilled	Memory crash	Pod killed due to RAM limit
📉 Resource Request	Minimum resource	Guaranteed CPU/memory
📈 Resource Limit	Max resource	Prevents overuse

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🎚️ QoS Class	Priority level (Quality of Service (QoS) classes)	to decide which Pods to evict from a Node experiencing Node Pressure.
⚠️ Node Pressure	Resource shortage	Memory, disk, PID pressure
🛡️ Pod Disruption Budget (PDB)	Availability rule	Limits pod down and ensuring high availability in a Kubernetes during node maintenance, cluster upgrades
👑 Leader Election	Master selection	Used by controllers
🧹 Garbage Collection	Auto cleanup	Deletes unused resources
🧩 Custom Controller	Custom logic	Watches CRDs
🤖 Operator Pattern	App automation	DB lifecycle management
🔵 Blue-Green Deployment	Zero-risk release	Switch traffic instantly
🟡 Canary Deployment	Gradual rollout	Test with few users
🔧 Node Drain	Safe node maintenance	Moves pods before shutdown
🚫 Cordoning	Block scheduling	Prevent new pods on node
📦 Static Pod	Node-level pod	Managed by kubelet only
🪞 Mirror Pod	API reflection	Static pod visible in API
🔐 RBAC	Access control	Who can do what (define roles and permissions for users, groups, or service accounts.)
🎯 Role	Namespace permission	Limited access
🌍 ClusterRole	Cluster-wide access	Admin-level permissions
🧾 ServiceAccount	Pod identity Used for API access (non-human account)	ServiceAccount is an identity used by applications (Pods) to talk securely to the Kubernetes API or other external services.
🔑 Token	Authentication key	Used by pods (token is a credential used to authenticate to the Kubernetes API server.)

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📊 Audit Log	Activity log (CCTV for your cluster API 🔍)	Audit Logs record WHO did WHAT, WHEN, WHERE, and RESULT for every request made to the API Server.
📍 NodeSelector	Simple placement	Run pod on specific node (✅ "Run this Pod only on nodes with these labels.")

📘 AWS EKS (Elastic Kubernetes Service) Glossary

Term	What It Is	Simple Explanation / Real Use
☁️ Amazon Web Services	Cloud platform	Provides infrastructure & managed services
🌐 Amazon EKS	Managed Kubernetes	AWS manages control plane
📦 EKS Cluster	Kubernetes cluster	Runs Kubernetes on AWS
🧠 Control Plane	Managed by AWS	API server, etcd, scheduler
🖥️ Worker Node	EC2 instance	Runs application pods
⚙️ Managed Node Group	AWS-managed nodes	AWS handles upgrades
🔧 Self-Managed Node Group	User-managed nodes	Full control over EC2
🚀 Fargate	Serverless compute	Run pods without EC2
🧩 EKS Add-ons	Managed components	CoreDNS, kube-proxy
🔗 Cluster Endpoint	API endpoint	kubectl connects here
🔐 IAM Role	AWS identity	Controls access to AWS
🧾 IAM Role for Service Account (IRSA)	Pod-level IAM	Secure AWS access for pods
📄 aws-auth ConfigMap	Auth mapping	Maps IAM users to RBAC
🔑 OIDC Provider	Identity provider	Enables IRSA
🌍 VPC	Network boundary	Isolates EKS cluster
📡 Subnets	Network segments	Public / private nodes
🛡️ Security Group	Firewall rules	Controls traffic

Term	What It Is	Simple Explanation / Real Use
🔌 ENI	Elastic Network Interface	Pod networking
🌐 VPC CNI	AWS network plugin	Assigns VPC IPs to pods
📍 Pod IP	VPC IP	Pods get real AWS IP
🔗 Service IP	Virtual IP	Internal cluster access
🌐 LoadBalancer Service	AWS ELB	Creates ALB/NLB
🚪 ALB Ingress Controller	Ingress handler	Uses AWS ALB
⚡ NLB	Network Load Balancer	High performance
💾 EBS CSI Driver	Block storage	Persistent volumes
📂 EFS CSI Driver	Shared storage	Multiple pods share files
📦 PersistentVolume (PV)	Actual storage	Backed by EBS/EFS
📄 PersistentVolumeClaim (PVC)	Storage request	Pod asks for storage
📈 Cluster Autoscaler	Node scaling	Adds/removes nodes
📊 HPA	Pod autoscaling	CPU/memory based
⚡ Karpenter	Smart autoscaler	Fast node provisioning
📊 CloudWatch Logs	Logging	Centralized logs
📈 Container Insights	Monitoring	CPU, memory metrics
📦 ECR	Container registry	Stores Docker images
🔄 EKS Upgrade	Version update	AWS upgrades control plane
💻 Node AMI	OS image	Used by worker nodes
🛠️ eksctl	CLI tool	Creates EKS clusters
🧑‍💻 kubectl	Kubernetes CLI	Manage resources
📦 Helm on EKS	Package manager	App deployments
🔒 Private Cluster	No public endpoint	Secure cluster access
🌍 Public Endpoint	Internet accessible	Easier access
🔐 Endpoint Access Control	Security setting	Public / private / both
🌐 Multi-AZ	High availability	Nodes across AZs
💰 EKS Pricing	Cluster cost	Per-hour cluster fee